only those vibrations in a direction perpendicular to the direction of molecular chains are transmitted. Thus, the light transmitted through the polaroid sheet is polarized. A sheet fabricated according to this process is known as H-sheet.

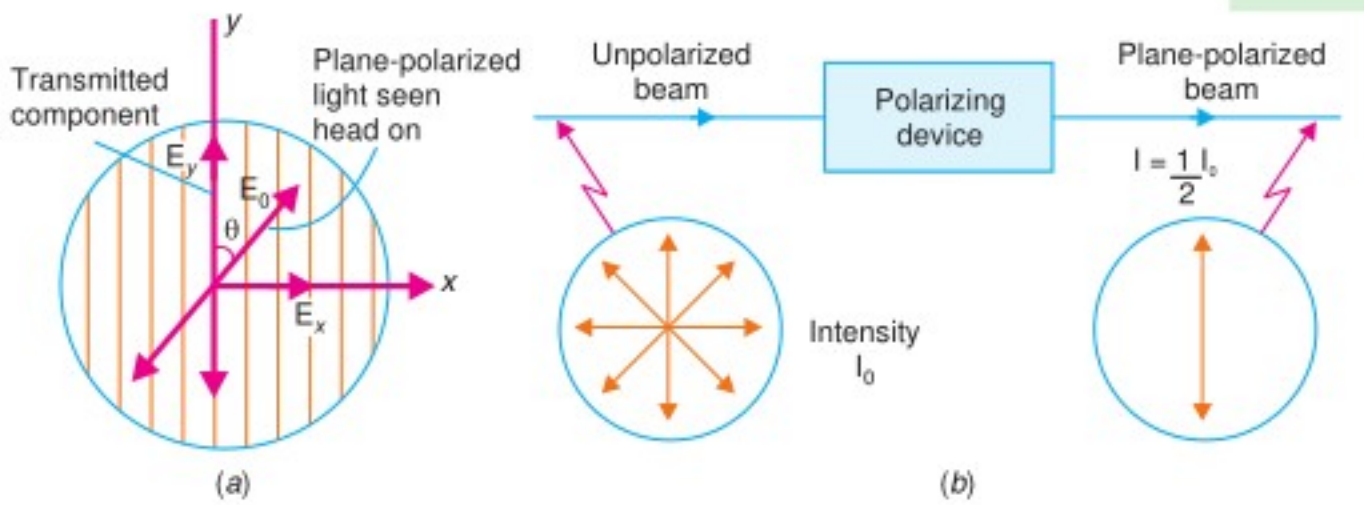
These sheet polarizers are inexpensive and can be made in large sizes. Polaroid sheets are widely used in sunglasses, camera filters etc to eliminate the unwanted glare from objects.

Polaroid sheets are extensively used for the production and detection of linearly polarized light.

20.8. POLARIZER AND ANALYZER

A *polarizer* is an optical element, which utilizes the phenomenon of selective absorption or double refraction, and transforms unpolarized light into polarized light. Plane polarized light is obtained by eliminating one of the two components in the unpolarized light. When natural light is incident on a polarizer, the *E*-field component that is parallel to the chains of iodine atoms induces current in the conducting chains and is therefore strongly absorbed. Consequently, the light transmitted contains only the component that is perpendicular to the direction of molecular chains. The direction of *E*-vector in the transmitted beam corresponds to the **transmission axis** of the Polaroid sheet.

Effect of polarizer on natural light



(*a*) Action of polarizer on linearly polarized wave
(*b*) The intensity of an unpolarized beam reduces to half after passing through a polarizer.

Fig. 20.20

Let us now understand the action of polarizer on the incident unpolarized light. Let us consider unpolarized light incident on a polarizer with the electric vector E_0 making an angle θ with respect to the *transmission axis* of the polarizer. The electric vector E_0 may be resolved into its component vectors lying parallel and perpendicular to the transmission axis of the polarizer (see Fig.20.20 *a*), that is E_y, parallel to the transmission axis and E_z, perpendicular to the transmission axis of the polarizer. The polarizer transmits the parallel component while blocking the perpendicular component. Thus, it is the parallel component E_y that is transmitted by the polarizer. But

$$E_y = E_0 \cos \theta \quad \ldots (20.7)$$

and hence, the intensity of the transmitted component is given by

$$I \propto E_y^2 = E_0^2 \cos^2 \theta \quad \ldots (20.8)$$

In unpolarized light all the values of θ are equally probable. Therefore, the fraction of light transmitted through the polarizer equals the average value of $\cos^2 \theta$.

$$\therefore \quad I = E_0(\cos^2\theta) = I_0(\cos^2\theta)$$

$$= \frac{I_0}{2\pi}\int_0^{2\pi}\cos^2\theta\, d\theta$$

$$= \frac{I_0}{2\pi}\int_0^{2\pi}\left(\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$= \frac{I_0}{4\pi}\left[(\theta)_0^{2\pi} + \left\{\frac{\sin 2\theta}{2}\right\}_0^{2\pi}\right]$$

$$= \frac{I_0}{4\pi}(2\pi + 0)$$

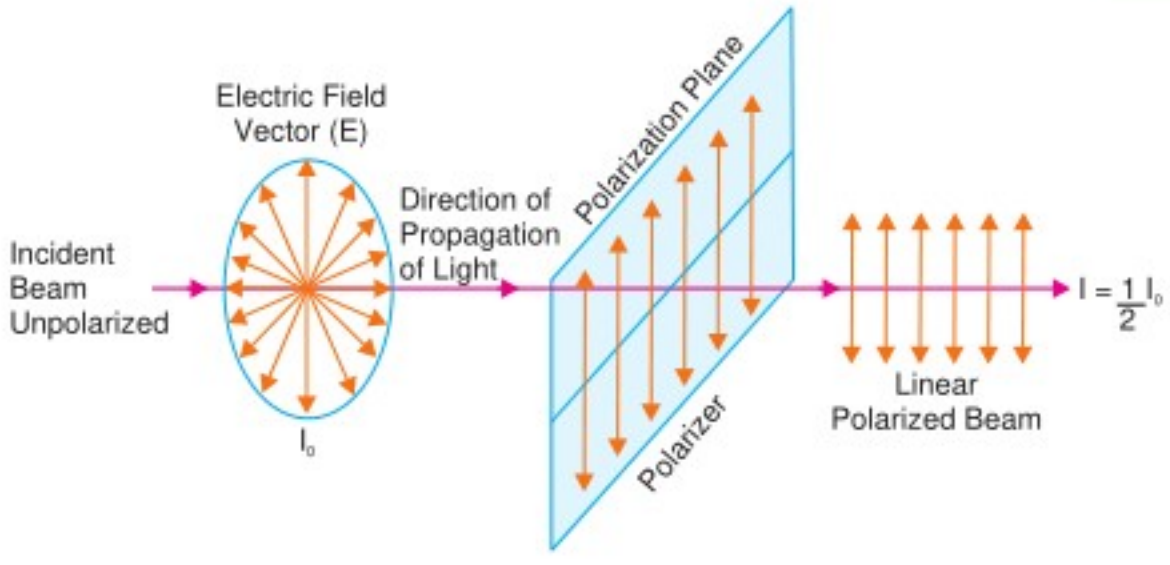
$$= \frac{1}{2}I_0$$

$$\therefore \quad I = \frac{1}{2}E_0^2 = \frac{1}{2}I_0 \qquad \ldots (20.9)$$

Thus, **if unpolarized light of intensity I_0 is incident on a polarizer, the intensity of light transmitted through the polarizer is $I_0/2$.**

20.8.1 PRODUCTION OF LINEARLY POLARIZED LIGHT USING A POLARIZER

A polarizer is associated with a specific direction called the **transmission axis** of the polarizer. If natural light is incident on a polarizer, only those vibrations that are *parallel* to the transmission axis are allowed through the polarizer whereas the vibrations that are in perpendicular directions are totally blocked. Therefore, the transmitted light contains waves oscillating in the same plane, as illustrated in Fig.20.21. Thus, the transmitted beam is linearly polarized.



Production of linearly polarized light

Fig.20.21

According to equ.(20.9), *when unpolarized light of intensity I_0 is incident on a polarizer, the intensity of light transmitted by the polarizer is $I_0/2$.*

20.8.2 DETECTION OF LINEARLY POLARIZED LIGHT

An *analyzer* is an optical element, which is used to identify the plane of vibration of plane polarized light.

To examine light coming from some direction either after emission or reflection etc, we use a Polaroid sheet. The Polaroid sheet used to determine the plane of polarization of light is known as an **analyzer**. There is no difference between a polarizer and analyzer in fabrication but they differ in their roles. Both the polarizer and analyzer are characterized by a transmission axis.

When the transmission axis of the analyzer A is set up parallel to that of polarizer P, light transmitted by the polarizer, passes unhindered through the analyzer (Fig. 20.22 *a*).

If the transmission axes are set at an angle q, light is partially transmitted (Fig.20.22 *b*). As the angle rotates from 0 to 90 degrees, the amount of light that is transmitted decreases. When the axes are perpendicular to each other, the polarized light from P is extinguished by the analyzer A (Fig.20.22 *c*). The polarizer and analyzer are said to be **crossed** in this configuration.

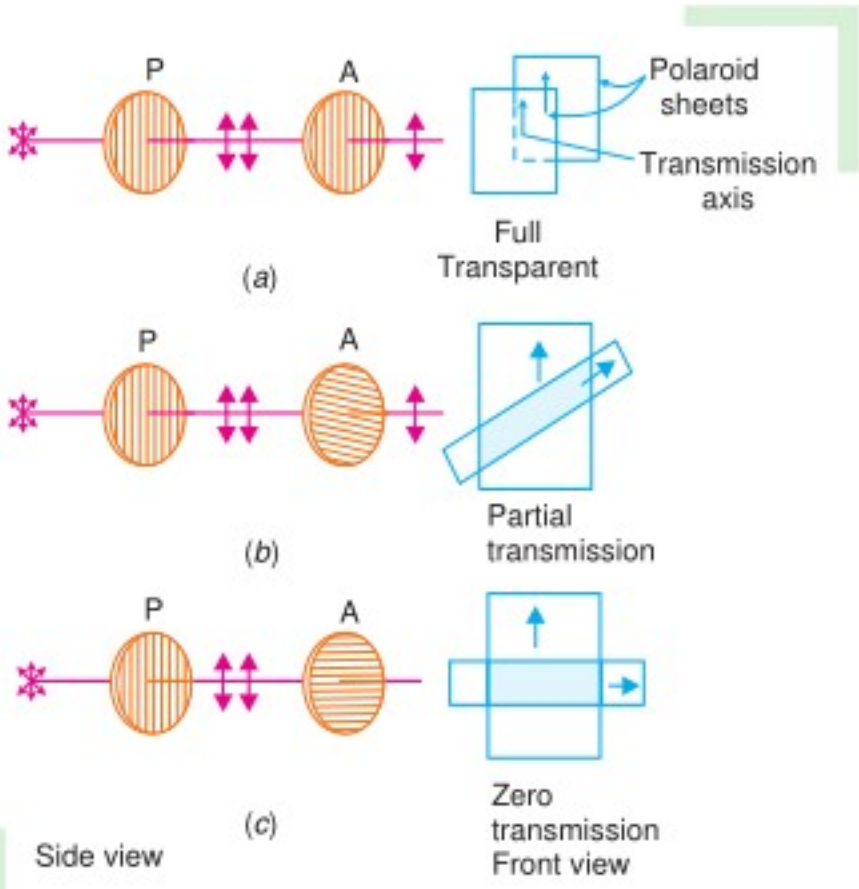


Fig. 20.22

When we rotate the axis of the analyzer with respect to that of the polarizer, we obtain two positions of maximum intensity and two positions of zero intensity in *one full rotation*. Conversely, if we obtain two positions of maximum intensity and two positions of zero intensity in one full rotation of the analyzer, we conclude that the light incident on the analyzer is plane-polarized light.

20.9. MALUS' LAW

The amount of light transmitted through a polarizer at an arbitrary angle [Fig.20.22 (*b*)] is given by **Malus's Law**.

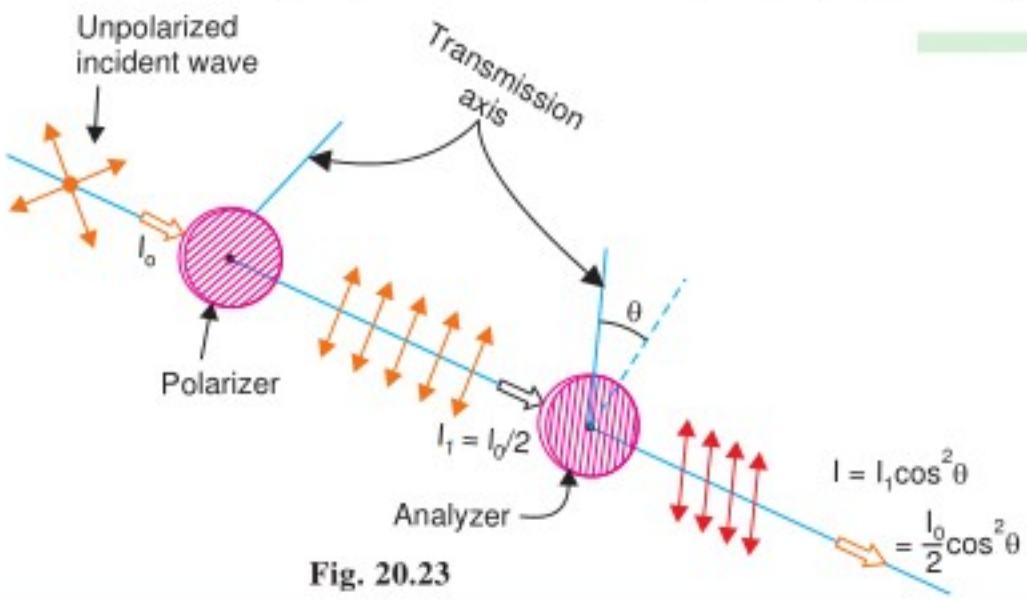


Fig. 20.23

In 1809 Malus found that the *intensity of polarized light transmitted through a polarizer is proportional to the square of cosine of the angle between the plane of polarization of the light and the transmisson axis of the polarizer.*

This statement is known as Malus' law.

If unpolarized light of intensity I_o is incident on a polarizer, plane polarized light of intensity $I_o/2$ is transmitted by it. Let us denote $I_o/2$ by I_1. Let this plane polarized light pass through an analyzer. The intensity of the light transmitted through the analyzer is given by

$$I = E_1^2 \cos^2\theta = I_1 \cos^2\theta = \frac{1}{2} I_0 \cos^2\theta \quad \ldots (20.10)$$

Light transmitted through the analyzer at specific settings are as follows.

Case (*i*): If $\theta = 0°$	axes parallel	$I = I_1 = \frac{1}{2} I_0$
Case (*ii*): If $\theta = 90°$	axes perpendicular	$I = 0$
Case (*i*): If $\theta = 180°$	axes parallel	$I = I_1 = \frac{1}{2} I_0$
Case (*i*): If $\theta = 270°$	axes perpendicular	$I = 0$

Thus, we obtain two positions of maximum intensity and two positions of zero intensity when we rotate the axis of the analyzer with respect to that of the polarizer.

20.10. ANISOTROPIC CRYSTALS

When a light beam is incident on an *isotropic medium* such as a glass slab, it refracts as a single ray. An optically isotropic material is one in which the index of refraction is the same in all directions. Glass, water and air are examples of isotropic materials. The atoms in a crystal are arranged in a regular periodic manner. If the arrangement of atoms differ in different directions within a crystal, then the physical properties vary with the direction. The thermal conductivity, electrical conductivity, velocity of light and hence refractive index etc properties depend on the crystallographic direction along which the property is measured. Then we say that the crystal is **anisotropic**. In such anisotropic crystals the force of interaction between the electron cloud and the lattice is different in different crystallographic directions. The natural frequency of the electron cloud is likewise dependent on the direction in which the electrons are caused to vibrate by the incident light wave. This results in different velocities in different directions and the index of refraction is different in different directions within the crystal.

The anisotropic crystals are divided into two classes: uniaxial and biaxial crystals. In case of **uniaxial crystals**, one of the refracted rays is an ordinary ray and the other is an extraordinary ray. In **biaxial crystals** both the refracted rays are extraordinary rays. Calcite, tourmaline and quartz are examples of uniaxial crystals whereas mica, topaz and aragonite are examples of biaxial crystals.

20.10.1 CALCITE CRYSTAL

Calcite is a common naturally occurring substance. Both marble and limestone are made up of many small calcite crystals bonded together. A large crystal of calcite is colourless and transparent. It was at one time found in great quantities in Iceland and hence it is also known as **Iceland spar**. Naturally occurring calcite crystals (Fig.20.24*a*) has rhombohedral cleavage which means it breaks into blocks with parallelogram-shaped faces. It is bounded by six faces (Fig.20.24*b*), each of which is a parallelogram with angles equal to 101°55'and 78°5'. The rhombohedron has only two corners A and H where all the face angles are obtuse (101°55'). These two corners appear as the *blunt corners* of the crystal. At the rest of six corners there is one obtuse angle and two acute angles.

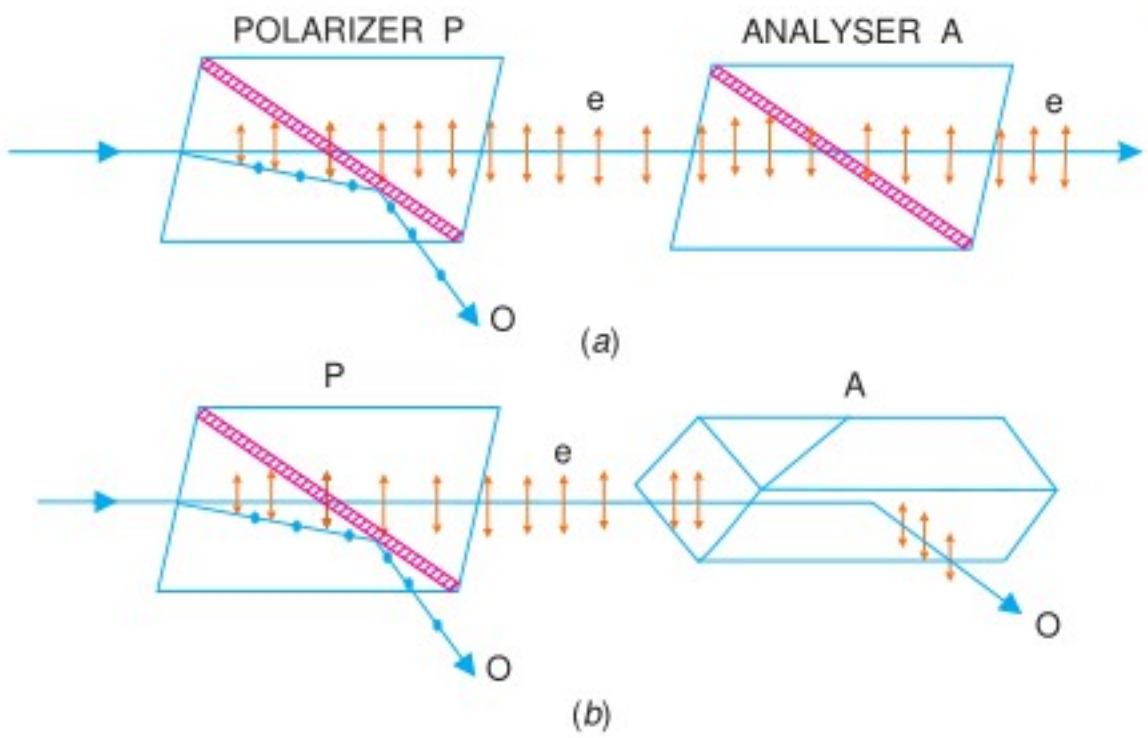


Fig. 20.30

P (Fig.20.30 b), the vibrations of the ray, emerging from P and incident on A, will be perpendicular to the principal section of A. In this position the ray behaves as o-ray inside the prism A and is totally internally reflected by the Canada balsam layer. Hence no light is transmitted by the prism A. In this configuration, the two Nicol prisms P and A are said to be **crossed.** If the Nicol prism A is further rotated through another 90°, the intensity of light emerging from A will go on increasing. The intensity will become a maximum when its principal section is again parallel to that of the prism P. Thus, the prism P produces linearly polarised light while the prism A detects it. Hence the prism P is called a polarizer and the prism A an analyser.

20.13. EFFECT OF POLARIZER ON LIGHT OF DIFFERENT POLARIZATIONS

The action of a polarizer, whether a Nicol prism or a Polaroid sheet, on light of different types of polarization is as follows:

(*i*) If **unpolarized light** is incident on a polarizer, it transmits half the intensity of light incident on it. The transmitted light stays constant in intensity on rotation of the polarizer.

(*ii*) If **partially polarized light** is incident on the polarizer, the intensity of the transmitted light depends on the direction of the transmission axis of the polarizer. The intensity varies from a maximum value I_{max} to a minimum value I_{min}. Two positions of I_{max} and two positions of I_{min} occur in one full rotation of the polarizer.

(*iii*) If **plane polarized light** is incident on the polarizer, the intensity of the transmitted light varies from zero to a maximum twice in one full rotation of the polarizer.

(*iv*) When **circularly polarized light** is incident on the polarizer, the transmitted light remains constant in intensity on rotation of the polarizer. This is interpreted as follows: the circular vibrations may be resolved into two mutually perpendicular linear vibrations of equal amplitude. When the circularly polarized light is incident on the polarizer, the vibration parallel to its transmission axis passes through the polarizer while the perpendicular component is not allowed. This happens in all positions of the polarizer in its rotation. Therefore, the intensity of the transmitted light stays constant in one full rotation of the polarizer.

(*v*) When **elliptically polarized light** is incident on the polarizer, the transmitted light varies in intensity from a maximum value I_{max} to a minimum value I_{min} on rotation of the polarizer (Fig.20.31). occurs when the transmission axis of the polarizer coincides with the semi-major axis of the ellipse and occurs when the transmission axis coincides with the semi-minor axis of the ellipse.

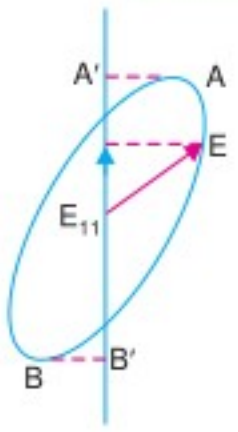


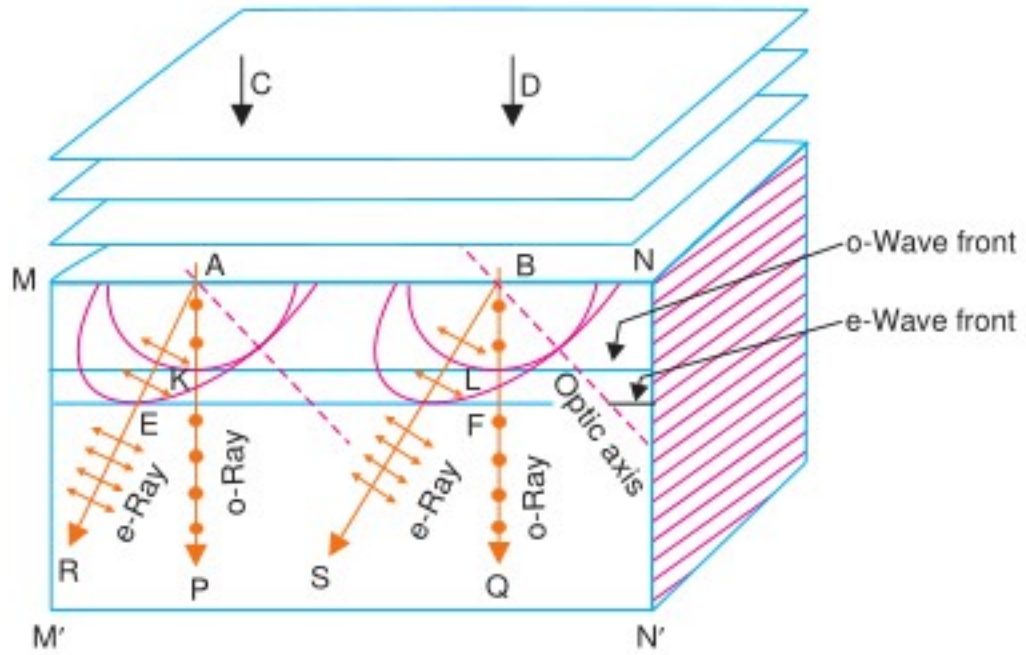
Fig. 20.31

20.14. HUYGENS' CONSTRUCTION OF WAVEFRONTS:

A thinner rectangular cross-section of a crystal can be cut from a bigger crystal in three different ways. It can be cut in such a way that the optic axis lies (*i*) inclined to the refracting face or (*ii*) parallel to the refracting face or (*iii*) perpendicular to the refracting face. The path of o-ray and e-ray within a uniaxial crystal can be determined using Huygens' principle of secondary wavelets. We take the example of a negative crystal for the purpose of tracing the paths of o-ray and e-ray.

Case.1. Optic axis inclined to the refracting edge:

Let MNN′M′ represent the calcite (negative) crystal. Let CD represent the monochromatic plane wave front incident normal to the crystal surface. The optic axis is ***inclined to the refracting edge.*** According to Huygens the phenomenon of double refraction involves two types of propagation of light waves. The wave known as o-wave has a spherical wave front whereas the e-wave has an ellipsoidal wave front. The two wave fronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.



Unpolarized light beam is incident at normal on a calcite crystal slice. Optic axis is inclined to the incident ray. Huygens wavelets showing double refraction, and difference in velocities of *o*-ray and *e*-ray.

Fig. 20.32

Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wave front becomes a source of secondary disturbance. The points A and B are chosen for the purpose of illustration.